YILDIZ TECHNICAL UNIVERSITY

APPLIED SQL (MTM4692)
PROJECT MILESTONE

Project Name: Multi-Exchange Crypto Arbitrage Detector
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GitHub Repository Link: https://github.com/ozergulcan/Crypto_Arbitrage_Detector.git

INTRODUCTION

This project aims to develop an SQL-based system to detect arbitrage opportunities in cryptocurrency markets. Arbitrage refers to making profit from price differences of the same asset across different exchanges.

In cryptocurrency markets, prices may vary across platforms such as Binance, Coinbase, Bitfinex, and Bitstamp. These differences may create short-term profit opportunities. In this project, historical price data from multiple exchanges is collected, cleaned, and analyzed using SQL.

PROBLEM STATEMENT

There are several challenges in detecting arbitrage opportunities:

- Different exchanges provide data in different formats
- Time formats and structures may vary across datasets
- Price differences alone do not guarantee profit due to transaction costs

Because of these issues, simply comparing prices is not sufficient. This project focuses on identifying **real arbitrage opportunities** by considering both price differences and transaction fees.

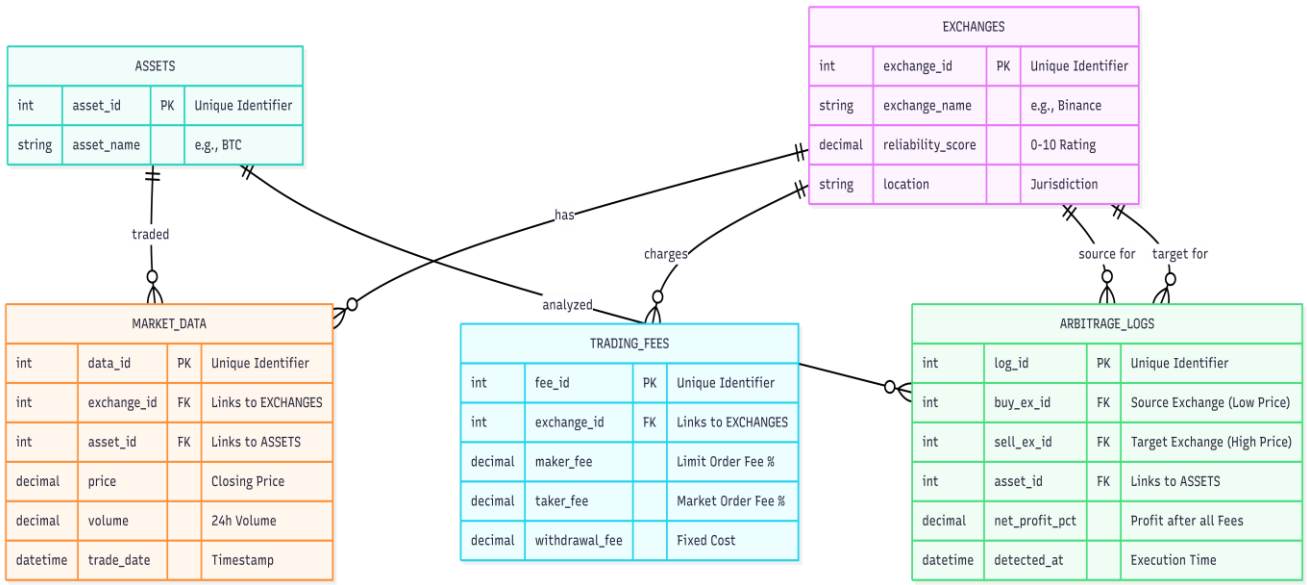
IMPORTANCE OF THE PROJECT

This project is important because it shows how SQL can be used for both data management and financial analysis. It also shows how real-world financial problems can be approached using structured data.

Additionally, working with multiple datasets helps to understand data cleaning, transformation, and integration processes.

DATABASE DESIGN

The system is designed using a relational database structure. It includes core tables and staging tables for data preprocessing.



Staging Tables

Separate staging tables are used for each dataset (Binance, Coinbase, Bitfinex, Bitstamp). These tables are used to import raw CSV data before transformation.

Relationships Between Tables

- Exchanges → MarketData (One-to-Many)
- Assets → MarketData (One-to-Many)
- Exchanges → TradingFees (One-to-One)

This structure allows efficient comparison of prices across exchanges.

SQL FEATURES USED

The following SQL features will be used in the project:

- CREATE TABLE
- PRIMARY KEY / FOREIGN KEY
- INSERT, UPDATE, DELETE
- SELECT queries
- JOIN operations
- GROUP BY and HAVING
- Aggregate functions (MIN, MAX) ,

- VIEWS & CTEs

These features are used to manage data and perform arbitrage analysis.

SAMPLE QUERIES

Query 1 – Data Integration : This query transfers cleaned data from staging tables into the main MarketData table.

```
INSERT INTO MarketData (exchange_id, asset_id, price, volume, trade_date)
SELECT 1, 1, Close, Volume, Date
FROM Binance_Staging;
```

Query 2 – Price Spread Detection: This query finds price differences between exchanges.

```
SELECT
   trade_date,
    MIN(price) AS min_price,
    MAX(price) AS max_price,
    MAX(price) - MIN(price) AS spread
FROM MarketData
GROUP BY trade_date
HAVING COUNT(DISTINCT exchange_id) > 1
```

Query 3 – Arbitrage Detection Using View: This query identifies potential arbitrage opportunities by comparing prices across different exchanges on the same date.

```
CREATE VIEW ArbitrageView AS
SELECT
   m1.trade_date,
    m1.exchange_id AS buy_exchange,
    m2.exchange_id AS sell_exchange,
    m1.price AS buy_price,
    m2.price AS sell_price,
    ((m2.price - m1.price) / m1.price * 100) AS gross_profit_pct
FROM MarketData m1
JOIN MarketData m2
   ON m1.trade_date = m2.trade_date
    AND m1.exchange_id <> m2.exchange_id
WHERE m2.price > m1.price;
```

CONCLUSION

In this project, an SQL-based system was developed to analyze cryptocurrency price data from multiple exchanges and detect arbitrage opportunities. By combining data from different sources and standardizing it into a single structure, it became possible to compare prices effectively.

Overall, this project provides a simple but effective approach to understanding arbitrage in cryptocurrency markets and demonstrates the importance of clean and well-structured data.